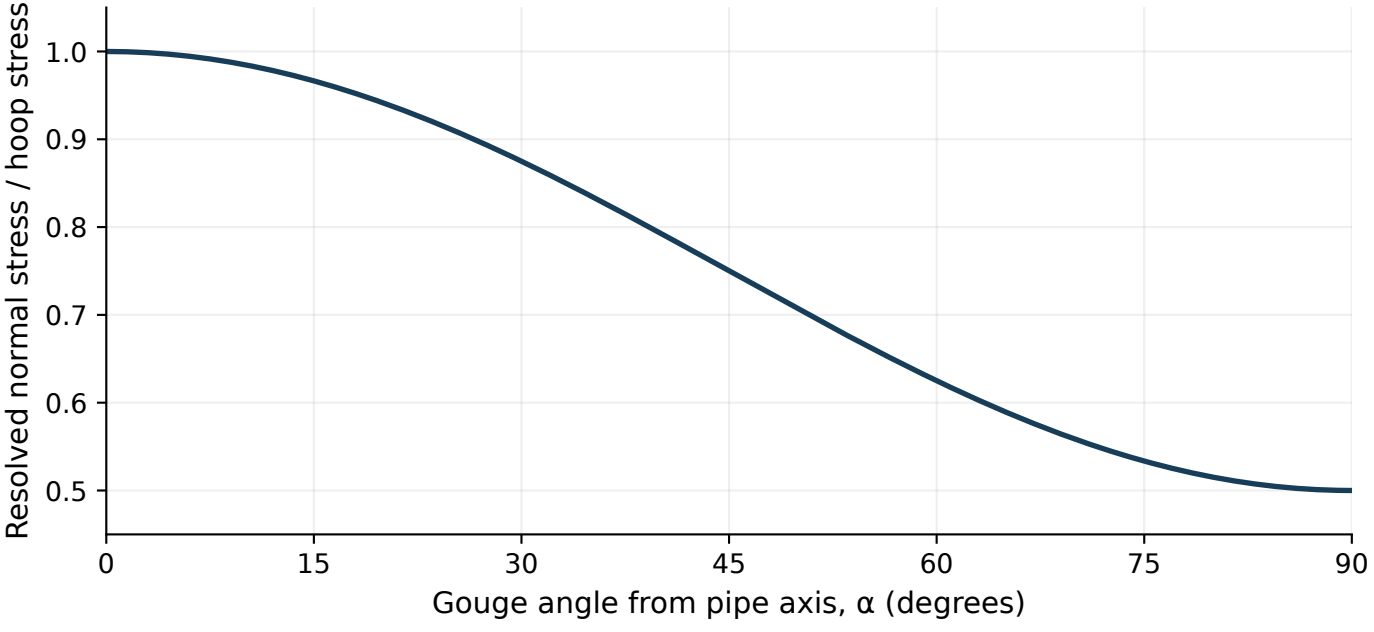
Analytical intact-membrane reference
Resolved normal stress / hoop stress
1.0
0.9
0.8
0.7
0.6
0.5
0
15
30
45
60
75
90
Gouge angle from pipe axis, α (degrees)